Chapter 4: Does money matter? Some evidence for the importance of monetary policy

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1 Overview

Most newspapers pay a great deal of attention to every utterance of the Chairman of the Federal Reserve. Moreover, the stock market often gyrates wildly based upon perceptions of how the Federal Reserve is expected to move at any given time. Overall, there seems to be general consensus in the popular press that monetary policy plays a fundamental role in shaping economic outcomes. For many outsides, then, it may come as a surprise that there is no such consensus within the economic profession. For example, Edward Prescott, winner of the 2004 Nobel Prize in Economics, noted that "Postal economics is more central to understanding the economy than monetary economics" and "it is an established scientific fact that monetary policy has had virtually no effect on output and employment in the US since the formation of the Fed." In response to the Federal Reserve quantitative easing measures Prescott opined "Bond buying is as effective in bringing prosperity as rain dancing is in bringing rain."

At some level, I suspect that this skepticism, which I think is deeply rooted in the profession, stems from the fact that, at an intellectual level, *it is* mysterious that monetary policy has any affect *in principle*. After all, the nominal price of a good is simply a measurement unit. Roughly speaking, the more the nominal stock of money, the higher the price (even if we will refine that statement considerably later on!). Why should the unit of measurement matter? Does anybody think that the game of football would drastically change if each down were not measured in 10 yards, but instead 9.144 meters? Or that the results of the world championship in 100 meter sprint would be any different if it were measured instead by 328 feet? In the same vein, should it matter, in principle, if milk costs 1 dollar or 100, as long as in the latter case people have 100 times as many nominal units of money?

Notwithstanding this skepticism, I find myself in sync with the popular press and conventional wisdom about the importance of monetary policy. But before

moving further it seems worthwhile to summarize *why* I hold that view. Or to put it more broadly, *why* do the majority of professional economists hold this view?

This chapter has roots in a question I got from a colleague, who is a professor at University of Pennsylvania, some years ago when we were both visiting Princeton. He asked, "Why are you so convinced that monetary policy matters?" I was a bit stumped in response, and I realized that I had just been taken it as given for a long while, just as the popular media does. While I do not exactly recall the response I gave (I think I may have mumbled something about the Great Depression), following this question and in preparation for a graduate class I was about to teach at Yale I started asking professional economists the same question and taking notes on the answer. Roughly, this chapter reports the results of this informal survey. I think the answers I got can be summarized as follows, roughly in order of importance:

1. Time Series Narrative
2. Vector Autoregression (VAR)
3. Event Studies of Monetary Policy
4. Micro-data
5. International Prices
6. Wall Street's Response to Monetary Policy

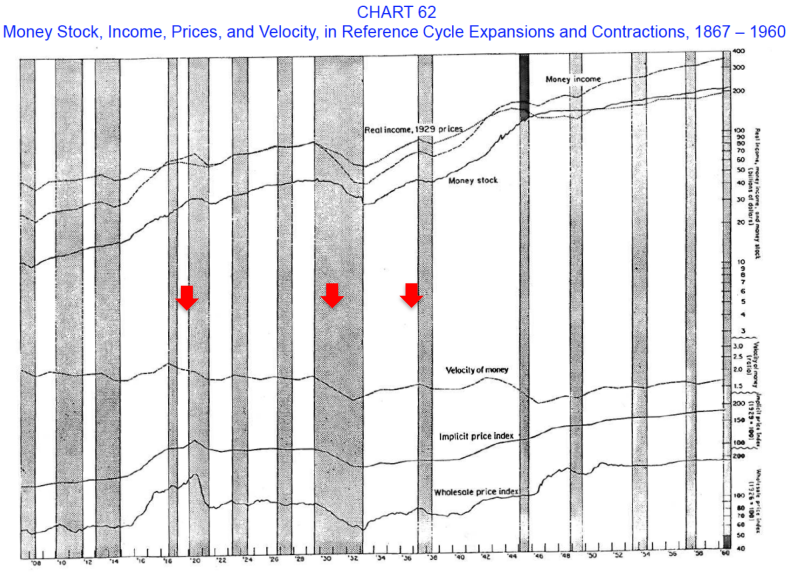
2 Time Series Evidence: Narrative

If you ask economists of certain generation (the baby boomers in particular, think of, for example, people of the generation of Ben Bernanke and Christina Romer) they will invariably bring up that the work of Friedman and Schwartz (1963) [4] convinced them of the importance of monetary policy. Given how frequent this response is, here I briefly summarize this evidence and then illustrate the evidence from Romer and Romer (1989) [7], which is very much in the same spirit and is, in addition, a special favorite of mine. Also, this evidence is very useful in understanding the important innovations of the VAR evidence.

2.1 Friedman and Schwartz (1963)

Friedman and Schwartz, in their 1963 book *A Monetary History of the United States* [4], looked at specific episodes in which monetary policy, particularly

Figure 1: They key idea of Friedman and Schwartz



changes in the volume of the money supply, had impacts on the larger economy. Friedman and Schwartz isolated four crucial episodes they coined "natural experiments" that they argued stand out in the monetary record and that "like the crucial experiments of the physical scientist, the results are so consistent and sharp as to leave little doubt about their interpretation." These "experiments" correspond to January – June 1920, October 1931, and July 1936 – January 1937. They also noted that the massive decline in the money stock as a result of Fed inaction from 1929 – 1933 could be regarded as a fourth crucial experiment.

The figure shows Friedman and Schwartz's compilation of the money supply over this period and in particular highlights the three "experiments" associated with the contraction in the money supply there is a decline in output.

The first main innovation in Friedman and Schwartz was their compilation of a time series for different measures of the money stock and their argument that sustained decline in it triggered recession, though prior to the book the prevailing wisdom had been that monetary policy played a secondary role in explaining the Great Depression. The second main innovation was the systematic use of qualitative information from primary sources about monetary policy. In particular, they argued in a way many found compelling that the "natural experiments" that they argued resulted in considerable contractions were the result of unforced errors by monetary policy makers, and could have been avoided

had the Fed prevented collapse in the money supply.

While their work had a large impact, notable criticism includes that their their definition of money was vague and imprecise. As for the "natural experiments," one criticism is that the criteria for them is not obvious. Another common criticism is that movement in the money supply was not completely independent of the economic fluctuations Friedman and Schwartz were trying to explain. Moreover, they also lacked statistical tests to prove the significance of their described relation. Perhaps most fundamentally, however, the book did not provide any analytic framework to support the basic conclusion.

2.2 Romer and Romer (1989)

In 1989, Romer and Romer [7] did a natural followup of Friedman and Schwartz. They did so by reading Federal Open Market Committee (FOMC) minutes, looking for time periods in which the Federal Reserve decided to raise interest rate to bring down inflation. These have been termed the Romer dates. They are: October 1947, September 1955, December 1968, April 1974, August 1978, October 1979, and December 1988.

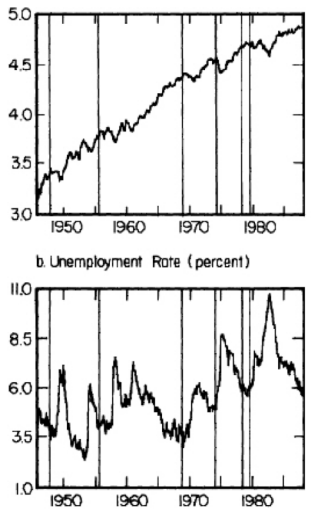
Figure 2.2 shows the Romer dates and plots them with monthly industrial production and the unemployment rate. What I especially like about this work is its simplicity and transparency. The way I like to think of the Romers' thought experiment is as follows: Consider a time period in which the Federal Reserve decides to raise interest rate with the explicit aim of bringing down inflation. If it were true that money was completely neutral, would it not be somewhat surprising if output would always contract immediately after the Fed tries to bring down inflation? As we see in the figure, industrial production typically declines after the Romer dates, and moreover, unemployment increases.

The Romer and Romer work is a logical continuation of the work of Friedman and Schwartz, but suffered from many of the same pitfalls in methodology. Most importantly:

- It is entirely possible that the policy tightening itself was a response to some external stimulus that also resulted in the decline in output.
- The choice of dates is highly subjective.

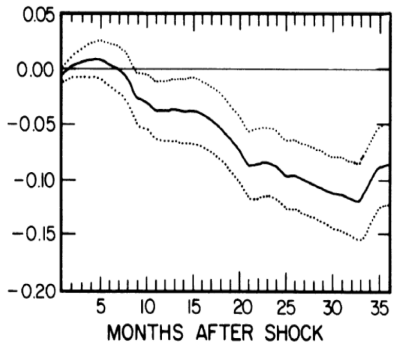
With respect to the first bullet point, consider the possibility that inflation always increases in response to a negative oil price shock and that the Federal Reserve tightens policy in response. In this case, the Romer dates may just be measuring the negative output effect of oil price shocks rather than the effect of a monetary tightening.

Figure 1 ECONOMIC ACTIVITY AND MONETARY SHOCKS.



Notes: Vertical lines are drawn at the dates of monetary shocks. The actual dates are October 1947, September 1955, December 1968, April 1974, August 1978, and October 1979. The sources of the data are described in the text. The data have been seasonally adjusted by a regression on monthly dummy variables.

Figure 4 IMPULSE RESPONSE FUNCTION FOR BASIC INDUSTRIAL PRODUCTION REGRESSION



3 Time series evidence: VAR Literature

The use of Vector Autoregression to understand the effect of monetary policy by Christopher Sims [8] was highlighted by the Nobel Committee as a key rationale for awarding him the Nobel Prize in 2011. One way of thinking about this evidence is that it offers a formal way of identifying monetary policy shocks that directly addresses some of the concerns one may have with the narrative evidence we just reviewed.

3.1 Christiano, Eichenbaum, and Evans (1999)

Christiano, Eichenbaum, and Evans (1999) [2] offer a good (even if slightly dated) overview of the VAR literature. A key equation in the VAR is an empirical characterization of policy given by:

$$-S_t = f(\Omega_t) + \sigma_t \epsilon_t^s \tag{1}$$

where S_t is the policy instrument (e.g. the federal funds rate), f is a linear function representing a monetary policy rule, Ω_t is the information at time t, and $\sigma_t \epsilon_t^s$ is some random shock to the system.

In this case, we are interested in the effects of a shock to the ϵ_t^s term.

The equation above is an example of a policy rule. A common example is the Taylor Rule:

$$i_t = \bar{\imath} + \phi_\pi(\pi_t - \pi*) + \phi_y Y_t + \epsilon_t^s \tag{2}$$

where $\bar{i}$ is the average interest rate, π_t inflation, $\pi*$ the inflation target and Y_t output. The parameters $\phi_y > 0$ and $\phi_\pi > 1$ are coefficients of the policy rule that determine how strongly the central bank responds to inflation being above or below the target – thus the central bank raises interest rate more than one to one if inflation is above the target and lowers it if it is below.

ϵ_t^s is a monetary policy shock which is purely exogenous. The first part of the policy rule is termed the systematic component of the policy rule – while the shock is its non-systematic component.

We can see what a prescribed Taylor Rule policy looks like over time in the figure below, from John Taylor's original article suggesting this rule. The point of the figure was that it seemed to do a decent job in characterizing policy under Chairman Greenspan:

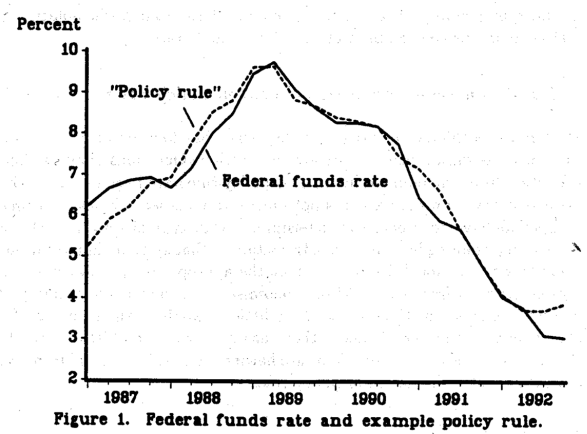


Figure 1. Federal funds rate and example policy rule.

3.2 Notes on VAR's

3.2.1 Monetary Policy Shocks in VAR Models

What is a monetary policy shock in the policy rule specified above, e.g. the Taylor rule? There are a few possible interpretations:

- Preference shocks (such as a new voting member on the FOMC or political pressures)
- Measurement errors at the time of decision-making that could obscure the optimal choice
- Stochastic dimension of choice, for example, lab experiments suggest that humans choices are not fully predictable and include an element of chance.

Note: A contractionary shock according to these criteria is different from the Romer and Romer interpretation of monetary tightening. In their case, they looked for situations when the Fed decided to bring down inflation even at the cost of output contraction. That would be captured by the term $\phi_\pi(\pi_t - \pi*)$ in the Taylor rule, sometimes referred to as the systematic part of the policy rule. The monetary shock, in contrast, is the part of interest rate variations that cannot be explained by the difference in inflation from its target or an output gap.

To take the oil price shock we mentioned before as an example, suppose there is an oil price shock that increases inflation. This will then be picked up by the systematic component of the policy rule – not the shock. Thus the VAR offer a possible response to the criticism of the Narrative approach.

3.2.2 The VAR(q) Model

A VAR(q) model is represented by a system of equations in the form below:

$$Z_t = B_1 Z_{t-1} + \cdots + B_q Z_{t-q} + u_t \tag{3}$$

where Z_t is a vector of dimension k of variables of interest. Often included are output, inflation, and the federal funds rate. u_t is one-step-ahead forecast error, and it is uncorrelated with variables dated t-1 and earlier (in practice researchers add enough lags of Z for this to be the case).

Even if we know the B's and the vector u_t, we still can't identify the values of ϵ_t, which correspond to the structural shocks. The elements of the vector u_t, which contains the forecast error of each variable, will typically be correlated with one another.

3.2.3 Identification: Sims (1980) and CEE (1999)

In order to identify these structural shocks, we assume that ϵ_t is a vector of uncorrelated "fundamental shocks," and that $A_0 u_t = \epsilon_t$. We will need to make some additional assumptions in order to back out and identify ϵ_t.

The most common assumption made in the literature is that the matrix C $= A_0^{-1}$ is lower triangular (Cholesky), more specifically:

$$u_t = C\epsilon_t = A_0^{-1}\epsilon_t$$

Then, this implies that

$$A_0 Z_t = A_1 Z_{t-1} + \cdots + A_q Z_{t-q} + \epsilon_t \tag{4}$$

This Cholesky assumption implies the following: that the first forecast error u_t^1 is the first fundamental shock, and that the second forecast error is the part of u_t^2 that is not correlated with u_t^1, and the same holds for each subsequent choice of n.

3.2.4 CEE (1999)

Consider the following autoregression:

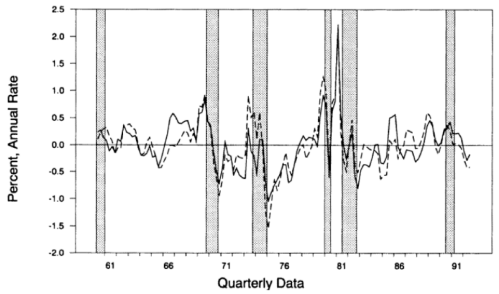
$$Z_t = \begin{bmatrix} GDP_t \\ Inflation_t \\ FederalFundsRate_t \end{bmatrix}$$

If one uses the Cholesky decomposition on this vector the assumption is that the Federal Funds Fate can respond to GDP and Inflation, but GDP and Inflation cannot respond to the Federal Funds Rate.

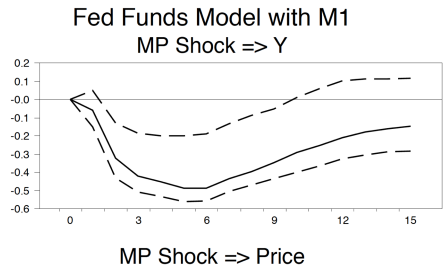
The following two figures are from CEE (1999) [2]. The first shows monetary policy shocks, with the shaded areas illustrating NBER recession. The point of the figure is that each of the shaded regions seem to be preceded by a monetary policy shock.

The second figure is the impulse response of output to a monetary policy shock. The point of the figure is that output contracts after a negative monetary policy shock, with the effect peaking after around six months. This impulse response function is typical for the VAR monetary literature.

FIGURE 1. — THREE QUARTER, CENTERED AVERAGE OF FF POLICY SHOCKS WITH AND WITHOUT COMMODITY PRICES



Fed Funds Model with M1



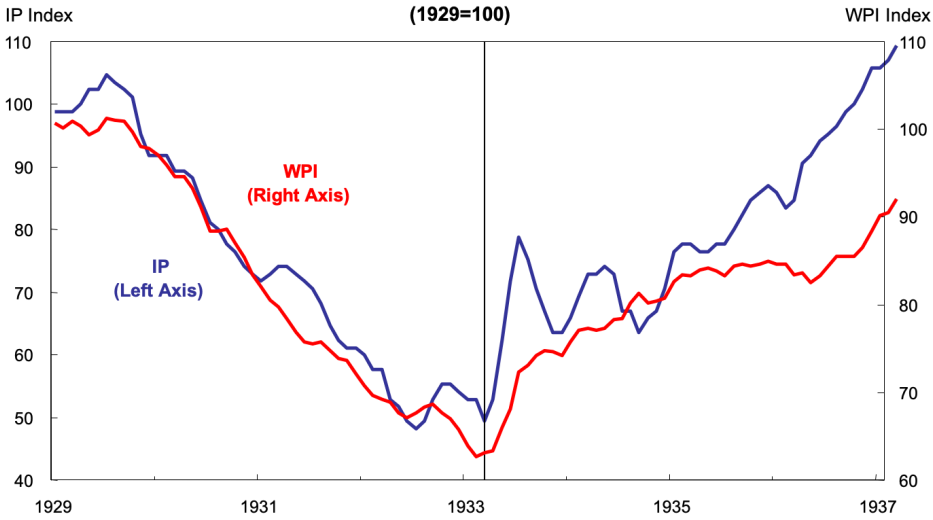
4 Event Studies

One common answer from economists, when asked for evidence in favor of the effect of monetary policy, is to point to particular historical episodes that are difficult to reconcile with monetary non-neutrality. Here I summarize a few such episodes.

4.1 The Great Depression

4.1.1 FDR and Reflation

Franklin Delano Roosevelt went off the gold standard in 1933. This gave the Fed the freedom to inflate the price level, which it did. We discuss this in better detail later in the class, but the figure below shows that once FDR took power (the vertical line) both output and inflation rebounded very strongly. Eggertsson (2008) discusses this episode in detail. Eggertsson and Pugsley (2006) discuss another interesting episode, the recession of 1937, which can be interpreted through the same lens, as well as the recovery of 1938. In 1937 FDR abandoned his commitment to inflate, and then once again recommitted to inflation in 1938.



Source: Federal Reserve Board, NBER Macrohistory Database

4.1.2 Eichengreen and Sachs (1985)

The US was not the only country that went off gold in response to the Great Depression. Eichengreen and Sachs [3] examined the fates of countries that abandoned the gold standard, and especially their recoveries from the Great Depression.

The figure below shows the industrial production and exchange rates of various countries in 1935 in comparison with their value 1929. Clearly, those countries that inflated their currency by uncoupling themselves from the gold standard ended up recovering faster and doing better than those that remained tied to the gold standard, and the more so, the more they were willing to inflate.

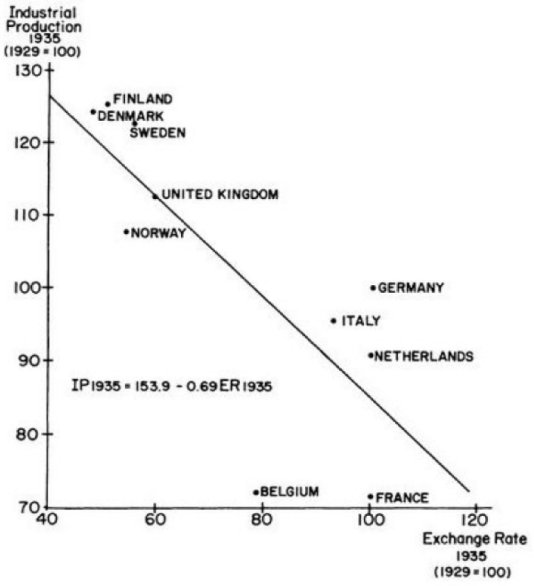


FIGURE 1
CHANGES IN EXCHANGE RATES AND INDUSTRIAL PRODUCTION, 1929–1935

The above examples illustrate the power of monetary policy in helping countries recover from the worst economic crisis in modern history.

4.2 Eighteenth-Century France

In 1724, France conducted a curious monetary experiment documented by Velde (2009) [10]. Below you see a picture of a coin in France from the time.



The curious thing about this coin, or livre, was that it did not have a number on it. So if a glass of beer cost 1 livre, you would have no idea how many coins to use without me giving you further clarification. It was the government that decided by decree how many livres each coin was worth.

And now back to the curious experiment documented by Velde. One day the French government simply said that coins were worth 40 percent less in livres. This, of course, led to a dramatically reduced the purchasing power of people holding these coins, at least as measured by units of livres.

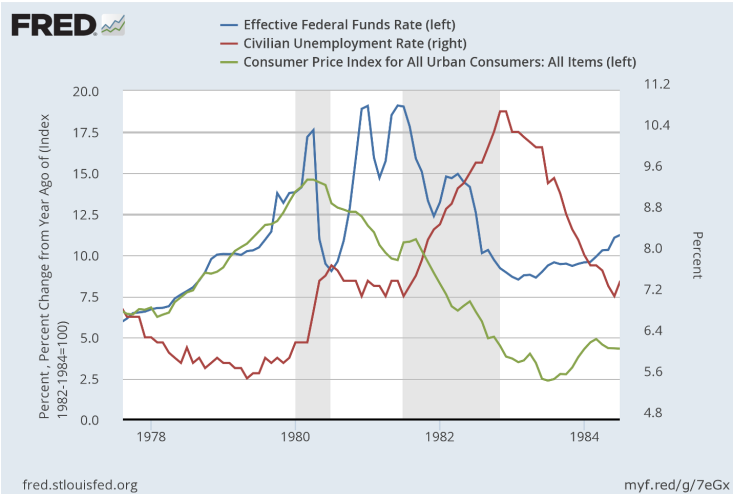
This reduction of purchasing power depressed the economy in a variety of ways. Because coins were worth less, more coins were required to pay off debts, because the debt contracts were written in livres, not number of coins. As a result, more coins were required to pay off debts. In addition, it took time for the prices of goods to adjust, because again, prices of goods were measured in livres, not number of coins. This led to a decline in GDP. In fact, goods prices never completely corrected. These changes ultimately led to a 30 percent contraction in French industrial production.

Notably, several years later, the French government increased the value of the coins by about 20 percent. Then there was a large rebound in prices and production.

4.3 The Volcker Recession

During the 1970s, the United States was plagued by stagflation, a period of high inflation and unemployment. In order to curb it, in August 1979, Jimmy Carter selected Paul Volcker as the Chairman of the Federal Reserve. His job was to bring it down.

Volcker and the FOMC decided that it was time to act. As a result, they increased interest rates dramatically. This triggered a recession. The below chart shows Volcker's battle against inflation over his term:



As we can see, initial attempts to reduce inflation failed, so Volcker then more than doubled interest rates, which triggered a recession, but we can ultimately see that the commitment to high interest rates succeeded in dramatically bringing down inflation. But it also seems clear there was an output cost of this, manifested by the rise in unemployment.

5 Microdata

Here is very stylized account of many monetary models: Suppose that when you wake up in the morning, your money is worth twice what it was yesterday. As a result, you can now purchase twice as much as you could yesterday. This is because prices do not adjust immediately, and because you are spending more, output rises as a result. Over time, prices adjust, but the presumption is that prices do not adjust instantaneously.

One way to look for evidence if monetary policy works via this mechanism, i.e. that prices are sticky, is to check if this basic assumption of monetary models is, in fact, consistent with the underlying data. If it is true that prices are very rigid over periods of time policy is changing, it can – at best – be considered indirect evidence that monetary policy has output effects (even if one could in principle try to write down a model which is consistent with price rigidities – yet where money has no effect on output).

The idea of stickiness of prices is well-supported by the data. See the below chart of the price of saltines over time in Chicago supermarket, a figure taken from Nakamura and Steinsson (2008) [6]:

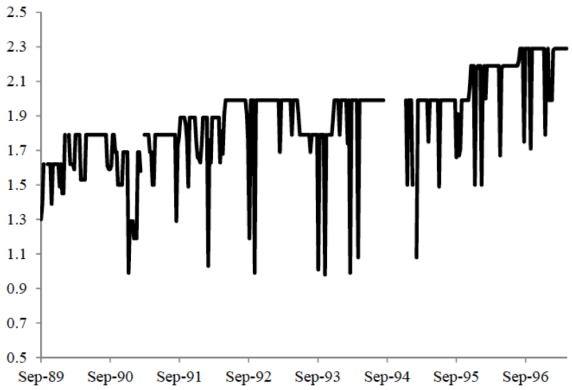


Figure 2
Price of Nabisco Premium Saltines 16 oz. at a Dominick's Finer Foods Store in Chicago

You notice that there are downward spikes in prices. These are sales. But you may also notice that, excluding sales, prices don't seem to be changing all

that frequently (and actually the sales price also seems to be recurrent). If you look at CPI as a whole, which is what Nakamura and Steinsson do, you find that the median duration of "regular" prices is about 8.5 months in the period 1988-1997, while it is about 11 months in the period 1997-2005.

Table 1: Frequency of Price Change in the CPI

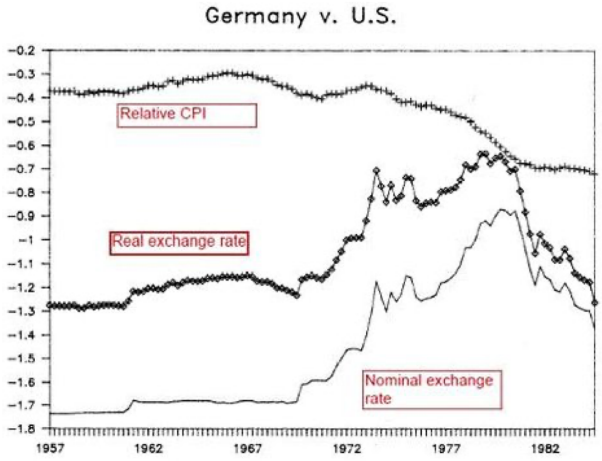
	1988-1997		1998-2005	
	Regular Price	Price	Regular Price	Price
Median Freq. of Change	11.1	20.3	8.7	19.4
Median Implied Duration	8.5	4.4	11.0	4.6
Median Freq. of Change Incl. Subs.	12.7	21.7	10.9	20.5
Mean Freq. of Change	18.7	23.9	21.1	26.5
Mean Implied Duration	11.6	8.3	13.0	9.0

6 International Prices

If all prices were flexible then relative prices such as the real exchange rate should only be determined by real factors as well. At the very least, the evolution of the real exchange rate should be independent of the monetary policy regime.

In the past, during the Bretton-Woods era, the nominal exchange rate was constant as a result of the Bretton-Woods agreement, which pegged all currencies to the dollar. You can see this in the figure below, which documents the nominal exchange rate between Germany and US. It is stable until the early 1970s when the US abandoned the Bretton-Woods agreement. During the fixed nominal exchange rate period, the real exchange rate is very stable as well.

As Bretton Woods disintegrates, the nominal exchange rate starts fluctuating. As the figure makes clear, however, the real exchange rate starts fluctuating more or less one-to-one with it as well. This is often referred to as the Mussa puzzle. Mussa suggested that a natural reason for this was that prices are very rigid. If they were not, he argued, then the behavior of the real exchange rate should not depend upon the monetary policy regime.



7 Wall Street

Examining Wall Street's response to possible changes in monetary policy is also an important piece of evidence. There is recent work that looks at high frequency movement in asset prices which is broadly supportive of an important role for monetary policy.

8 References

[1] Mark Bils and Peter J. Klenow. "Some Evidence on the Importance of Sticky Prices". *Journal of Political Economy* 112.5 (2004), pp. 947–985.

[2] Lawrence J. Christiano, Martin Eichenbaum, and Charles L. Evans. "Monetary Policy Shocks: What Have We Learned and to What End?" In: *Handbook of Macroeconomics*. Ed. by John B. Taylor and Michael Woodford. Vol. 1. Elsevier, 1999. Chap. 2, pp. 65–148.

[3] Barry Eichengreen and Jeffrey Sachs. "Exchange Rates and Economic Recovery in the 1930s". *The Journal of Economic History* 45.4 (1985), pp. 925–946.

[4] Milton Friedman and Anna J. Schwartz. *A Monetary History of the United States*. Princeton University Press, 1963.

[5] Jordi Galí. *Monetary Policy, Inflation, and the Business Cycle*. Princeton University Press, 2015.

[6] Emi Nakamura and Jón Steinsson. "Monetary Non-neutrality in a Multi-Sector Menu Cost Model". *The Quarterly Journal of Economics* 125.3 (2008), pp. 961–1013.

[7] Christina D. Romer and David H. Romer. *Does Monetary Policy Matter? A New Test in the Spirit of Friedman and Schwartz*. Tech. rep. 2966. NBER, 1989.

[8] Christopher A. Sims. "Macroeconomics and Reality". *Econometrica* 48.1 (1980), pp. 1–48.

[9] John B. Taylor. "Discretion versus policy rules in practice". *Carnegie-Rochester Conference Series on Public Policy* 39.1 (1993), pp. 195–214.

[10] François R. Velde. "Chronicle of a Deflation Unforetold". *Journal of Political Economy* 117.4 (2009), pp. 591–634.

[11] Michael Woodford. *Interest and Prices: Foundations of a Theory of Monetary Policy*. Princeton University Press, 2003.